Web Fundamentals

1. Web 3.0 does not have an exact definition, but there are arguments that it is already evolving, since it has abilities and dynamism that Web 2.0 did not have. For example, algorithms determine advertising using AI for user-specific content. Use of AI and cloud-based features are part of Web 3.0, but the article argues that the defining feature of Web 3.0 is that it is more user-centric and devolved agency more to users than to tech companies

2. Front-end:

The front-end of a website is the part that the users will see and interact with; in other words, the browser where the user interface appears. The front-end is also known as the presentation layer.

The front-end is built using three key components.

HTML (HyperText Markup Language): Every website out there is built using HTML. It's the backbone of the world wide web, used to define the site's structure and content.

CSS (Cascading Style Sheets): CSS will define how the HTML appears visually on a web page. CSS determines fonts, colours, layout, background images and more.

JavaScript: This programming language allows front-end developers to add more interactivity to the websites they build. Recent developments in JavaScript (and improvements in browsers) have made it possible to do more complex work on the front-end, creating features that used to be reserved for more complex back-end languages.

The back is organized more according to creativity than logic.

The back-end is the part of the website that users don't see. It refers to the server side, which is responsible for returning the page, and for storing or fetching user data. It's also known as the data access layer.

Back-end programming languages (like Python, PHP or Ruby) are often used to build content management systems and large, complex web applications.

3. Step one is the client request. The user's browser sends an HTTP request for the web-based server that hosts it. The response is then retrieved. This may include:

1. Informational responses (100–199): ○ 100 Continue

2. Successful responses (200–299): ○ 200 OK: The request succeeded, and the information was returned in the response.

3. Redirection responses (300–399): ○ 301 Moved Permanently: The requested object has been permanently moved, and the new URL is specified in the “Location” header of the response message. This will automatically be retrieved by the client.

4. Client error responses (400–499): ○ 400 Bad Request: Generic error code indicating that the request could not be understood by the server. ○ 404 Not Found: The requested document does not exist on this server.

5. Server error responses (500–599): ○ 505 HTTP Version Not Supported: The requested HTTP version is not supported by the server.

Languages like Javascript and CSS are used to deliver those responses through a browser, which acts as an interpreter.